

Subject to:

- i. $S_i P_{i0} + \sum_{j=1}^n x_{ij}(P_{i0} + MI(x_{ij})) = V_{\S}$
- ii. $S_i = \sum_{j=1}^n x_{ij}$
- iii. $r_{ij} = \sum_{k=1}^j x_{ik}$
- iv. $x_{ij} \geq 0$

The new objective function in Equation 10.16 correctly incorporates a variable trading cost function along with estimated return and portfolio risk. The decision variables are S and x_{ij} although S is computed from x_{ij} . We distinguish between these two variables for notation purposes only—the only true decision variable for the optimizer is x_{ij} . The important decision variable for portfolio managers is the number of shares to hold in the portfolio S , while the important decision variable for traders is how those shares need to be transacted over time.

The first constraint above ensures that the entire cash value $V_{\S}$ will be invested into the portfolio and follows from the definition of expected transaction value in Equation 10.9. The second constraint defines the number of shares that will be held in the portfolio. The third constraint defines the cumulative number of shares transacted in period j . The fourth constraint is an optional constraint and can be specified for a cash investment only $x_{ij} \geq 0$, liquidation only $x_{ij} \leq 0$, no constraint to incorporate both buys and sells.

Following Equation 10.16 the new portfolio optimization problem can be separated into the traditional investment and trading horizons as follows²:

$$\text{Max}_x \left(\sum_{i=1}^m (S_i P_{it} - S_i P_{i0}) - \lambda \cdot (t-n) S' C^* S \right) - \left(\sum_{i=1}^m \sum_{j=1}^n x_{ij} MI(x_{ij}) + \lambda r_j' C^* r_j \right) \quad (10.17)$$

This new formulation brings four interesting aspects to light. They are:

1. Risk aversion λ is the same for both portfolio manager and trader.
2. Trader's dilemma is not dependent upon any benchmark price.
3. Portfolio optimization with trading costs requires a multi-period process.
4. The portfolio manager's and trader's decisions are not separable—they are linked by $S = \sum x$.

²This separation of the portfolio optimization problem into corresponding trading and investment horizons was first presented publicly in December, 2003 expanding on the work of Kissell and Malamut presented in *Optimal Trading Strategies*.